

Step 3 of the scope-completion roadmap: the (D) diagonal-isotropy and (S)

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The probe of Steps 1–2 covered the off-diagonal sector (O). Step 3 confirms that the other two sectors of the Reading-H comparison decomposition — (D) diagonal isotropy and (S) selection floor — extend to the non-lattice competitor class. The conclusion is that both are *competitor-agnostic*: they depend on the layer G_* and the coherence packing bound, not on the competitor’s lattice structure, so they carry over unchanged. No tier change.

2. (D) Diagonal isotropy is competitor-agnostic

The second variation of the diagonal functional F_0 at the isotropic dressing G_* is (T-016 / hdiag-fullcovariance-formulation)

$$d^2F_0 = \frac{1}{2} \int G_*(q)^{-2} (dG)^2 d^3q + \Phi''_{\text{diag}} (dM_{\text{tot}})^2, \quad \Phi''_{\text{diag}} = \frac{3}{2} u_{\text{eff}}.$$

Decomposing an *arbitrary* perturbation $dG(q) = \sum_{lm} a_{lm}(|q|) Y_{lm}(\hat{q})$ and using $\int Y_{lm} d\Omega = 0$ for $l \geq 1$, the rotation-invariant interaction $\Phi_{\text{diag}}(M_{\text{tot}})$ curves ONLY the $l = 0$ sector, so the Hessian is block-diagonal:

$$l = 0 : H_0 = \frac{1}{2} G_*^{-2} + \frac{3}{2} u_{\text{eff}} > 0, \quad l \geq 1 : H_l = \frac{1}{2} G_*^{-2} \geq \frac{1}{2} r_R^2 = 4.64 \times 10^{-2} > 0,$$

the $l \geq 1$ infimum analytic at the gap shell $|q| = q_0$. **This decomposition never references the competitor’s lattice structure**: a non-lattice competitor is simply another anisotropic direction dG , seen by the SAME positive block-diagonal Hessian. The entropy floor $\frac{1}{2} r_R^2$ is rotation-invariant (G_* isotropic), so it bounds the curvature toward every angular direction equally. Hence the strict diagonal minimum $F_{\text{diag}}[Q] > F_{\text{diag}}[G_*]$ holds for EVERY competitor, lattice or not.

3. (S) Selection floor is competitor-agnostic

The SC-SCOPE selection joint (scscope_endpoint_sweep)

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + \text{composed}/S_{\text{un}} + R_{\text{max}} C_2}, \quad \rho = \frac{4(1 - a_0) \text{MARGIN}}{(\lambda I)^2 J_{\text{eff}} n_{\text{pack}}},$$

is built entirely from *layer* quantities — MARGIN (Prop-A layer margin), J_{eff} (minimum-transfer envelope), S_{un} (sunset cap), R_{max} (realized quartic) — and the coherence packing bound n_{pack} , which is the same for every admissible competitor. It carries no explicit dependence on the competitor’s additive (lattice) structure. Numerically $\text{joint} = 1.040 > 1$ at the operating point, and the structural saturation $\text{joint} \rightarrow S_{\text{un}} = 1.13$ as $\rho \rightarrow \infty$ is a layer property. Hence $\text{joint} > 1$ holds identically for non-lattice competitors.

4. The full (D)(O)(S) decomposition extends to C_full

Collecting the three sectors for the full admissible class $\mathcal{C}_{\text{full}}$:

sector	C_full status	margin
(D) diagonal isotropy	competitor-agnostic (this note)	comfortable ($\frac{1}{2}r_R^2, \frac{3}{2}u_{\text{eff}} > 0$)
(O) off-diagonal	thin closure (Step 2, res5_033)	thin ($\times 1.026$; EXT $\rightarrow \times 6$)
(S) selection floor	competitor-agnostic (this note)	comfortable (joint = 1.04, cap 1.13)

Thus the full Reading-H comparison decomposition holds for $\mathcal{C}_{\text{full}}$, the ONLY thin ingredient being the off-diagonal margin $\times 1.026$ (which the EXT conjecture would upgrade to $\times 6$). No B1/B2 tier change: this records the last structural ingredient of the C_full route; the residual items are the EXT margin upgrade (T2) and the operator decision on accepting the thin off-diagonal margin for a C_full statement.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Is (D) really lattice-free, or does the competitor enter the diagonal part elsewhere?” The diagonal functional depends on G only through $M_{\text{tot}} = \int G$ (rotation-invariant) and the per-mode entropy; the competitor’s angular positions enter only as the *direction* of dG , and every direction sees the positive block-diagonal Hessian. No lattice input.
- (β) “Is (S) truly competitor-agnostic, given the off-diagonal used K_{floor} ?” The off-diagonal (O) is the competitor-dependent sector (via K_{floor}); the SELECTION joint (S) is the third-cumulant LAYER balance (MARGIN/Jeff/ n_{pack} /SUNSET), which does not carry K_{floor} . The two are distinct sectors; (S) is agnostic.
- (γ) “Does this complete the C_full T7?” No — it shows (D)/(S) extend; the off-diagonal is thin (Step 2); the C_full statement still awaits the EXT margin upgrade and an operator decision on the thin margin, plus external referee. No tier change.

Quantitative sanity check (mandatory): (D) entropy floor $\frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$, breathing $\frac{3}{2}u_{\text{eff}} = 4.03 > 0$; (S) joint = 1.040 > 1, sunset cap 1.130; both competitor-agnostic. All from res5_034 (6/6).

6. Result footer

Result ID: B1/B2 (D)/(S) non-lattice extension (roadmap Step 3)

Precise statement: (D) diagonal isotropy (T-016): the second variation at G_* is block-diagonal in angular momentum with $l=0$ curvature $(1/2)G_*^{-2} + (3/2)u_{\text{eff}} > 0$ and $l \geq 1$ curvature $(1/2)G_*^{-2} \geq (1/2)r_R^2 = 4.64 \times 10^{-2} > 0$; the spherical-harmonic decomposition uses no lattice input, so the strict diagonal minimum holds for ANY competitor (lattice or non-lattice). (S) selection joint = MARGIN/(C2+composed/SUNSET+RMAX*C2) is built from layer + coherence-packing quantities (no competitor lattice structure), joint=1.040>1, sunset cap 1.13; holds identically for non-lattice. Both (D)/(S) competitor-agnostic => extend to C_full unchanged.

Scope: (D)/(S) sectors for C_full. With Step 2 (0 thin), the full (D)(0)(S) decomposition extends to C_full: (D)/(S) comfortable, (0) thin x1.026.

Dependencies: No B1/B2 tier change.
T-016/res5_016 (diagonal isotropy);
hdiag-fullcovariance-formulation;
scscope_endpoint_sweep (joint); res5_033 (Step 2 off-diagonal C_full).

Evidence grade: STRUCTURAL (competitor-agnosticism) + EXECUTED (res5_034 6/6).

Reproduction command: python
codes/vacuum/res5_034_DS_nonlattice_extension.py

Expected output: entropy floor 4.64e-2>0; breathing 4.03>0;
joint 1.040>1; sunset cap
1.130; 6/6 PASS.

Falsifier: A competitor entering the diagonal functional beyond M_tot + entropy;
or a selection-joint competitor-structure dependence beyond n_pack.

Tier before / after: No change. (D)/(S) competitor-agnostic (extend to C_full); with Step 2 the (D)(0)(S) decomposition holds for C_full (0 thin).

No-overclaim statement: Records (D)/(S) competitor-agnosticism (structural). Does NOT complete the C_full T7: the off-diagonal margin is thin (x1.026), and the C_full statement awaits EXT (margin upgrade) + operator thin-margin decision + external referee. No tier widening.

Next required action: Step 4 external referee; operator decision on thin off-diagonal margin; optional EXT promotion (margin upgrade). Then Step 5 consolidation.